Reproducible Sparse-Concept and Calibration Diagnostics for Knowledge Tracing

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Abstract*—Knowledge Tracing (KT) systems that skip, remediate, or advance practice typically consume a predicted probability rather than an area-under-the-curve (AUC) score, so population ranking can look acceptable while probabilities are poorly calibrated on rarely practiced knowledge components (KCs). This paper reports a diagnostic evaluation—not a new KT architecture—of train-only KC-frequency strata on three public logs (ASSISTments 2012, Junyi Academy, and XES3G5M) with IRT, DKT, and a Transformer KT baseline (T-KT). Lower KC training frequency does not universally degrade discrimination, but calibration can become less reliable in some sparse-concept regimes. On ASSISTments 2012, T-KT expected calibration error (ECE) rises from 0.114 on dense KCs to 0.228 on sparse KCs (Limited occupancy, N≈415); Junyi has no learner-based sparse stratum, and XES3G5M T-KT ECE is essentially flat. A locked global threshold at τ=0.7 raises the T-KT false-advance rate (incorrect-response rate among advance decisions) from 0.196 (dense) to 0.268 (sparse) on one ASSISTments fold; the sparse–dense gap stays positive on all five training runs and on all four unique partitions (mean 0.047). This is a simulated decision gate, not a classroom intervention. Probability-threshold decisions should be validated by KC-frequency stratum rather than on population AUC or ECE alone.

***Keywords*—*knowledge tracing, calibration, sparse concepts, learning analytics, educational decision support, mastery threshold*

I. INTRODUCTION

Adaptive practice platforms and intelligent tutoring systems use Knowledge Tracing (KT) to estimate a learner's command of a skill and to decide whether to skip, remediate, or advance practice [1]–[3]. Those systems rarely consume a population area-under-the-curve (AUC) statistic. They consume a predicted probability p, which is then compared with a threshold τ. If p≥τ, the platform typically withholds additional practice on that skill. The operational chain is therefore population AUC → predicted probability → calibration → threshold-based educational decisions, with particular diagnostic risk on sparsely trained knowledge components (KCs).

KT evaluations typically report population AUC and accuracy [4], [5]. Those metrics rank models, but they can conceal miscalibration on low-frequency KCs—skills with little training-fold evidence—because most test events lie on dense, frequently practiced concepts. A model can discriminate well in aggregate and still assign overconfident probabilities on the sparse tail. When a fixed threshold is applied to p, that overconfidence appears as an advance decision followed by an incorrect next response. We term this the false-advance rate (FAR). Section III defines FAR as P(y=0 | p≥τ); y denotes observed next-response correctness, not latent mastery truth.

This paper treats that mismatch as an evaluation problem for educational technology, not as a reason to propose another KT architecture. The study is organized around three research questions. RQ1: Does lower KC training frequency systematically degrade predictive discrimination? RQ2: How does calibration vary across KC-frequency strata and datasets? RQ3: When a fixed probability threshold is applied, does decision-error behavior differ between sparse and dense KCs?

The empirical answers are bounded. Lower KC training frequency does not universally degrade discrimination: on XES3G5M, sparse AUC is higher than dense AUC for DKT and T-KT. Calibration can, however, become less reliable in some sparse-concept regimes. On ASSISTments 2012, T-KT expected calibration error (ECE) increases from dense to sparse KCs, and a locked gate at τ=0.7 yields a higher FAR on sparse than on dense advances. The same ECE gradient is absent on Junyi, where the learner-based sparse stratum is empty, and is essentially absent for T-KT on XES3G5M. We therefore do not claim that sparse KCs always fail, and we do not claim a causal effect of training frequency on calibration.

Contributions are conservative: (i) a train-only KC-frequency protocol with an explicit strict cold-start group, so the definition of "sparse" cannot leak test-fold counts; (ii) per-stratum calibration (ECE and Brier decomposition) on three public datasets, with occupancy flags (Reliable / Limited / Insufficient); (iii) a locked-threshold simulation of FAR and miss rates, with a check of the sparse–dense FAR gap on ASSISTments 2012 across five training runs / four unique learner partitions. We do not propose a new KT architecture, a new calibration algorithm, or a new auditing theory, and we do not report a classroom intervention. This is an evaluation-systems contribution for threshold-based educational decisions, not a new pedagogy. A graph-KT (GKT) model and a CL4KT-style adapter appear only as an exploratory single-fold diagnostic on ASSISTments.

II. LITERATURE REVIEW

A. Knowledge Tracing and benchmark models

Knowledge Tracing (KT) models a learner's evolving proficiency from historical interactions in order to predict the next response [3]. Classical approaches include Bayesian Knowledge Tracing (BKT), which tracks a latent mastery state for each skill [1], and item-response theory (IRT), of which the one-parameter Rasch model is the canonical form used in this study [6]. Deep Knowledge Tracing (DKT) replaced hand-specified mastery dynamics with a recurrent sequence model and established the modern next-response prediction task [2]. Attention-based successors, including context-aware attentive KT [7] and published SimpleKT [4], retain that task while substituting self-attention for recurrence. Benchmarking libraries such as pyKT have standardized preprocessing and comparison, and have shown that evaluation choices can distort reported discrimination [5]. Across this line of work, the primary reported statistic remains population AUC. Uncertainty-aware models such as UKT represent knowledge states as distributions [21]; they are architecture proposals evaluated mainly on population ranking, not a train-only frequency-stratum decision audit.

B. Graph and self-supervised KT

Graph-based KT (GKT) represents knowledge components as nodes of a graph and updates proficiency with a graph neural network [8]. Contrastive learning for KT (CL4KT) trains representations from augmented views of learning histories [9]. These models are related architectures, not the contribution of the present paper. They are included later only as an exploratory single-fold diagnostic on ASSISTments 2012, not as a proposed method and not as a state-of-the-art comparison.

C. Sparse-data and cold-start problems

The term “sparse” is used in several incompatible senses in the KT literature. This paper distinguishes four. (1) Sparse attention: sparseKT applies k-sparse attention so that only a subset of historical interactions receives nonzero weight [16]. (2) Sparse KC frequency: a knowledge component with few training-fold observations. (3) New-student cold start: prediction for learners who have little or no interaction history at test time [17]. (4) Concept-level cold start: a knowledge component with zero training-fold frequency, regardless of whether the learner is new. Educational interaction logs additionally exhibit a long tail in which a small set of dense KCs dominates event volume [20].

These four problems are not interchangeable. sparseKT-style sparse attention is not equivalent to low-frequency KCs: attention can be sparsified on a frequently practiced skill. New-student cold start is not the same as concept-level zero-frequency cold start: a previously observed learner may still encounter a concept that never appeared in the training fold. The protocol in Section III operationalizes (2) and (4) through train-only KC-frequency strata with an explicit strict cold-start group (f=0).

D. Calibration and educational decision support

When predicted probabilities are consumed as educational decisions, ranking is not a sufficient evaluation [10]. Calibration asks whether the predicted probability p matches the empirical frequency of a correct next response. Expected calibration error (ECE) is the usual bin-wise summary of that mismatch [12]; modern neural networks can remain poorly calibrated even when they discriminate well [11]. The Brier score [13] and its reliability, resolution, and uncertainty decomposition [14] separate miscalibration from task difficulty, and reliability diagrams display the same probability-level comparison. Complementary post-hoc work has begun to correct per-item bias after a frozen KT backbone [15]. Probability-level evaluation of this kind is still typically reported on the pooled population, not on train-only frequency strata. Selective-prediction layers defer the most uncertain next-response scores to a teacher [22]. Abstention is complementary to a locked global threshold: this study does not abstain, and it reports the false-advance rate of the advances that a fixed τ would still make, stratified by train-only KC frequency.

Existing KT benchmarks mainly emphasize aggregate discrimination. Sparse frequency, calibration, sample support, and threshold-based decision error are rarely examined together. This study keeps standard models and reports those four quantities jointly, so that a population AUC result can be read as an educational-technology decision rather than only as a leaderboard rank.

III. MATERIALS AND METHODS

A. Datasets

We use three de-identified public logs after a common interaction schema: ASSISTments 2012 [18], [5], Junyi Academy [19], and XES3G5M [20]. Table 1 reports post-processing counts: processed learners, processed KCs, processed interactions, and learner-based test events. Exact processed interaction totals are 2,657,490 (ASSISTments 2012), 16,215,567 (Junyi Academy), and 6,413,353 (XES3G5M). Preprocessing drops rows with missing learner, KC, or label identifiers, binarizes correctness, parses timestamps, and keeps learners with at least two interactions. Those filters—not the raw public dumps—are what Table 1 counts.

Provenance. ASSISTments 2012 is the 2012–2013 public release with affect [18], documented for KT use in pyKT [5]; each retained row already carries one skill_id. Junyi Academy is the Kaggle Online Learning Activity Dataset [19] (not Junyi2015); the KC field is ucid. XES3G5M [20] is loaded from the authors’ kc_level split: train_valid_sequences.csv and test.csv are concatenated and flattened to one row per unmasked sequence position (padding and invalid labels dropped).

XES3G5M counts. The original dataset paper reports 18,066 students, 7,652 questions, 865 KCs, and 5,549,635 question-level interactions [20]. Table 1 lists 865 KCs and 6,413,353 valid KC-level rows: flattening still expands multi-KC questions into one row per listed concept, but padding tokens are excluded. Unique item_id is 7,652. Learner-based fold-0 test events are 1,282,422.

Table 1. Post-processing cohort statistics (learner-based split). Learner, knowledge-component (KC), interaction, and test-event counts are post-processing totals, not raw public-dump counts.

Dataset	Learners	KCs	Interactions	Test events
ASSISTments 2012	27,806	265	2.66M	534,150
Junyi Academy	71,014	1,326	16.2M	3,269,022
XES3G5M	18,066	865	6.41M	1,282,422

B. Splits and seeds

The learner-based split is primary. Users are shuffled with the fold seed; 20% of learners are held out as test, 10% as validation, and the remainder as training. Within a fold, train, validation, and test learners are disjoint. The temporal split is complementary: all interactions are sorted by timestamp; the earliest 70% are training, the next 10% validation, and the latest 20% test. Gate numbers below are not taken from the temporal split. KC-frequency buckets are constructed from the training file only.

Deep models are trained in five runs at seeds 42, 2024, 2025, 2026, and 2027 (fold_0 through fold_4). Seeds 2025 and 2026 share one student partition (fold_2 = fold_3) on all three datasets. These are five training runs and four unique student partitions, not five independent folds. Tables that report mean±sd therefore summarize four unique partitions: the two initializations on the duplicated split are averaged first. The gate-robustness table still lists all five training runs across four unique learner partitions.

C. Model settings

Main tables use local IRT, DKT, and a Transformer KT baseline (T-KT), not a pinned pyKT commit (training snapshot commit: NOT RECOVERED). The Transformer KT baseline is a two-layer Transformer encoder over DKT-style KC-response tokens, not the published SimpleKT architecture [4]. Table 2 reports recovered training settings; missing fields are marked NOT RECOVERED and are not imputed.

Table 2. Recovered training settings for the main baselines. NOT RECOVERED cells were not imputed.

Setting	**IRT 1PL**	**DKT**	**T-KT**
Implementation	local IRT	local DKT	local T-KT
Version/commit	NOT RECOVERED	NOT RECOVERED	NOT RECOVERED
Major hyperparameters	1PL: sigmoid(θ−β); L2 0.01	LSTM embed 64, hidden 128	Transformer 2 layers, 4 heads, embed 64
Optimizer	SGD	Adam	Adam
Learning rate	0.01	1e−3	1e−3
Early stopping	Fixed 10 epochs; final checkpoint.	Fixed 50 epochs; final checkpoint.	Fixed 50 epochs; final checkpoint.
Batch size	512	64	64
Max. sequence length	n/a	200	200
Selection metric	final checkpoint	final checkpoint	final checkpoint

On ASSISTments fold 0 only we also score a train-only GKT graph [8] (pyKT GKT; Adam, learning rate 1e−3, batch size 16, maximum length 100, hidden and embedding size 32, dropout 0.5, 20 epochs, patience 4, selection by validation AUC) and a CL4KT protocol adapter [9] (contrastive views on training sequences; not an official CL4KT checkpoint; Adam, learning rate 1e−3, batch size 64, maximum length 100, hidden size 64, 4 heads, 2 layers, dropout 0.2, 20 epochs, patience 6, selection by validation AUC).

In our implementation, unseen learners trigger a constant/base-rate fallback, yielding AUC=0.50; we report IRT as a base-rate reference, not as a ranking competitor. This fallback is not an inherent property of IRT.

D. Train-only frequency strata

For each fold, KC frequency f_{train} is counted on the training file only. Buckets: strict cold-start (f=0), very sparse (0<f<20), sparse (20≤f<100), medium (100≤f<500), dense (f≥500). Fig. 1 (top) shows the number of KCs in each stratum. Training-interaction volume is shown separately in the bottom row from summed f_{train}, not inferred from KC counts.

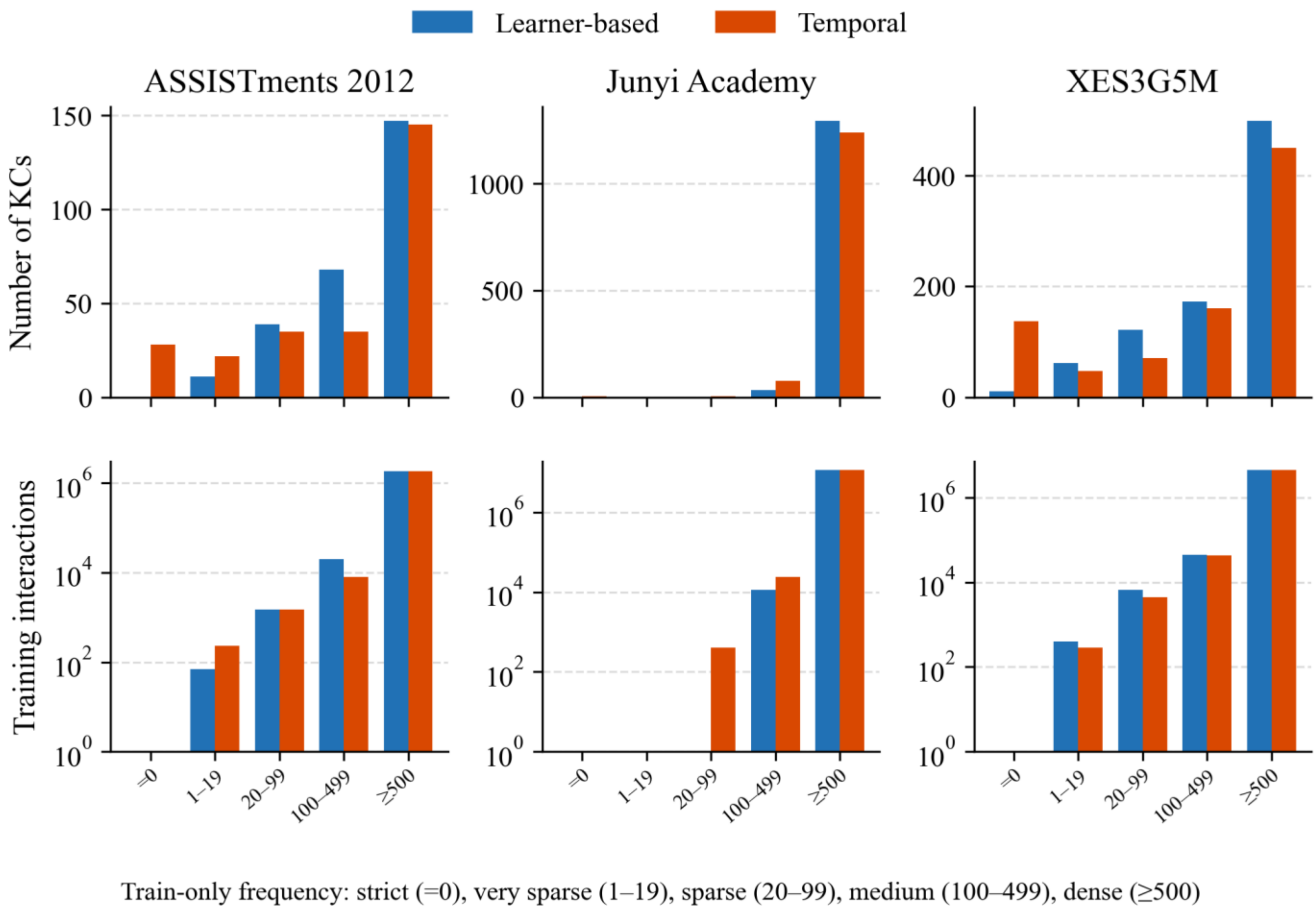

Fig. 1. Distribution of KCs across train-only frequency strata under learner-based and temporal splits. Bottom: verified training-interaction counts (sum of train-only f_{train} on fold 0); volume is not inferred from KC counts. Bottom y-axis is logarithmic.

E. Reliability flags

Occupancy flags follow test-event count N: Reliable (R) N≥1000; Limited (L) 100≤N<1000; Insufficient (I) N<100. We call R, L, and I descriptive sample-support flags. They are not inferential guarantees, confidence intervals, or hypothesis tests. Success claims require Limited or Reliable occupancy. Insufficient cells are descriptive only.

Very-sparse ASSISTments cells are Insufficient and descriptive only.

F. Calibration

Let y∈{0,1} be next-response correctness and p∈[0,1] the predicted probability. With M=15 equal-width bins, expected calibration error (ECE) is

$$ECE = \Sigma_m (n_m / N) |acc_m - conf_m| \quad (1)$$

The Brier score is (1/N) Σ_i (p_i − y_i)². We use the binned decomposition Brier = UNC − RES + REL [14], with UNC = ȳ(1−ȳ), REL = (1/N) Σ_m n_m (conf_m − acc_m)², and RES = (1/N) Σ_m n_m (acc_m − ȳ)². Because probabilities are binned, the three components are an empirical approximation and need not sum exactly to the directly computed Brier score. Lower ECE is better calibration; it is not a substitute for AUC.

G. Difficulty coupling

For each KC c we define a training-only difficulty proxy difficulty(c) = 1 − mean_train_correctness(c). The mean is taken on the training file only. This is an observational proxy for how often c is answered incorrectly in training; it is not a latent IRT difficulty parameter. The reported Spearman association is ρ(log(1+f_{train}), difficulty), computed on training-fold KC summaries. The same file yields item support (distinct items per KC), learner exposure (distinct learners per KC), and a curriculum-position proxy (median normalized sequence position). Results report between-KC associations and a within-KC sparsification check; neither identifies a causal frequency effect.

H. Simulated decision gate

For a locked global threshold τ, the system advances (skips remediation) if p≥τ and triggers remediation otherwise. Sparse events never receive their own tuned τ. The display threshold is τ=0.7; the grid {0.5, 0.6, 0.7, 0.8} is recorded in the artifact and is not a search over sparse error.

Let A={p≥τ}. The false-advance rate (FAR) is P(y=0 | p≥τ): among responses for which the simulated system would advance the learner, the proportion whose observed next response is incorrect. This is not a measurement of latent mastery. It is a next-response decision-error proxy under a simulated threshold gate. Miss = P(p≥τ | y=0). Expected FAR under the model probabilities is E[1−p | p≥τ]. Excess FAR = FAR − E[1−p | p≥τ]. ΔFAR = FAR_sparse − FAR_dense. FAR is a proportion of $N_{advance}$ = count(p≥τ) events, not of N_total; Miss is a proportion of $N_{incorrect}$ = count(y=0). N_total alone does not imply the precision of FAR. Where prediction-level exports exist, 95% intervals use the KC-clustered percentile bootstrap already used for this gate (B=2000, resample KCs within sparse and within dense). Positive ΔFAR means sparse advance decisions are less reliable than dense advance decisions. This is a simulated decision error, not an instructional randomized controlled trial (RCT).

IV. RESULTS

A. Aggregate discrimination

Table 3 reports overall learner-based area under the ROC curve (AUC) and accuracy (ACC), mean±sd over four unique partitions. This subsection is discrimination, not calibration. DKT is slightly above T-KT on all three datasets (ASSISTments DKT 0.6979±0.0014 versus T-KT 0.6837±0.0025). In our implementation, unseen learners trigger a constant/base-rate fallback, yielding AUC=0.5000; it is a base-rate reference, not a ranking competitor.

Table 3. Overall learner-based area under the ROC curve (AUC) and accuracy (ACC) (mean±sd over four unique partitions).

Dataset	Model	AUC	ACC
ASSISTments 2012	IRT	0.5000	0.6973±0.0004
ASSISTments 2012	DKT	0.6979±0.0014	0.7182±0.0014
ASSISTments 2012	T-KT	0.6837±0.0025	0.6996±0.0032
Junyi Academy	IRT	0.5000	0.7053±0.0018
Junyi Academy	DKT	0.7320±0.0009	0.7343±0.0015
Junyi Academy	T-KT	0.7231±0.0030	0.7274±0.0015
XES3G5M	IRT	0.5000	0.7961±0.0031
XES3G5M	DKT	0.8180±0.0009	0.8321±0.0015
XES3G5M	T-KT	0.7536±0.0010	0.8057±0.0029

On XES3G5M, sparse-stratum AUC is higher than dense-stratum AUC (DKT 0.858 versus 0.818; T-KT 0.847 versus 0.752; Reliable occupancy). Lower training frequency is therefore not a universal ranking failure. Junyi's learner-based sparse bucket is empty under the pre-specified cuts—a protocol outcome, not a missing cell and not a zero-ECE claim.

B. Calibration across frequency strata

Table 4 reports event-level expected calibration error (ECE), not AUC. On ASSISTments 2012, T-KT ECE increases from 0.1136±0.0066 (dense, Reliable, N=523,971) to 0.1541±0.0051 (medium) to 0.2280±0.0197 (sparse, Limited, N=415). That sparse cell is Limited support, not a high-N finding. DKT moves in the same direction on this dataset (dense 0.0602±0.0022 versus sparse 0.2333±0.0084). IRT dense ECE is 0.0031±0.0006, but resolution is zero (Supplementary Table S1): a near-constant predictor can look calibrated while providing no ranking information.

Calibration does not universally worsen. On Junyi Academy the learner-based sparse bucket is empty; only dense and medium exist, and T-KT ECE rises from 0.0792±0.0051 to 0.1073±0.0156. On XES3G5M, T-KT ECE is a counter-pattern: essentially flat from dense to sparse (0.1176, 0.1129, 0.1254) despite Reliable sparse occupancy (N=1,969). A flat ECE is not a license to skip occupancy reporting and is not the same as a flat miss rate. Full Brier, reliability, resolution, and uncertainty results are provided in Supplementary Table S1.

Table 4. T-KT event-level expected calibration error (ECE) by train-only frequency stratum. N is the mean test-event count across four unique learner partitions, not a single-partition count. Flags: Reliable (R) N≥1000; Limited (L) 100≤N<1000. Junyi sparse is empty.

Dataset	Stratum	N	Flag	ECE
ASSISTments 2012	dense	523,971	R	0.1136±0.0066
ASSISTments 2012	medium	5,963	R	0.1541±0.0051
ASSISTments 2012	sparse	415	L	0.2280±0.0197
Junyi Academy	dense	3,232,614	R	0.0792±0.0051
Junyi Academy	medium	3,836	R	0.1073±0.0156
XES3G5M	dense	1,269,345	R	0.1176±0.0014

Dataset	Stratum	N	Flag	ECE
XES3G5M	medium	12,889	R	0.1129±0.0061
XES3G5M	sparse	1,969	R	0.1254±0.0047

C. Threshold-based decision error

Table 5 is a simulated gate at τ=0.7 on ASSISTments 2012 fold 0 (seed 42), not a classroom trial. For T-KT, false-advance rate (FAR) is 0.196 [0.186, 0.208] on dense KCs (N=528,018; $N_{advance}$=284,326; $N_{incorrect}$=158,623; E[FAR]=0.113; Excess FAR=0.083; Miss=0.352) and 0.268 [0.202, 0.337] on sparse KCs (N=444, Limited; $N_{advance}$=235; $N_{incorrect}$=197; E[FAR]=0.050; Excess FAR=0.218; Miss=0.320). ΔFAR=+0.072. DKT FAR is 0.200 [0.190, 0.211] dense and 0.296 [0.221, 0.383] sparse. Exploratory GKT/CL4KT rows of the same table are discussed in subsection E, not as main-model findings.

Table 5. Simulated gate at τ=0.7, ASSISTments 2012 fold 0 (seed 42). FAR, Excess FAR, and Miss as defined in Section III.H. FAR 95% CIs: KC-cluster percentile, B=2000. Not a classroom trial.

Model	Stratum	N	$N_{advance}$	$N_{incorrect}$	FAR [95% CI]	E[FAR]	Excess FAR	Miss
T-KT	dense	528,018	284,326	158,623	0.196 [0.186–0.208]	0.113	0.083	0.352
T-KT	sparse	444	235	197	0.268 [0.202–0.337]	0.050	0.218	0.320
DKT	dense	528,018	315,650	158,623	0.200 [0.190–0.211]	0.147	0.053	0.398
DKT	sparse	444	243	197	0.296 [0.221–0.383]	0.057	0.239	0.365
GKT (train-only)	dense	528,018	351,503	158,623	0.205 [0.194–0.217]	0.163	0.042	0.455
GKT (train-only)	sparse	444	209	197	0.220 [0.149–0.295]	0.157	0.063	0.234
CL4KT (adapter)	dense	528,018	307,479	158,623	0.185 [0.176–0.194]	0.175	0.010	0.359
CL4KT (adapter)	sparse	444	200	197	0.240 [0.159–0.330]	0.116	0.124	0.244

Seed-42 ΔFAR 95% CI (KC-cluster, B=2000): T-KT [0.006, 0.138] (locked C2); DKT [0.019, 0.175]; GKT [−0.054, 0.092]; CL4KT [−0.018, 0.142].

Table 6 checks whether T-KT ΔFAR is a single-partition artifact. It is positive in 5/5 training runs (mean 0.047, sd 0.033) and positive in 4/4 unique partition-level estimates (mean 0.056, range 0.015–0.087). Mean sparse denominators are N=413, $N_{advance}$=227, $N_{incorrect}$=155. A KC-clustered bootstrap on seed 42 yields a 95% interval [0.006, 0.138] for ΔFAR—excluding 0, but wide, as Limited occupancy and $N_{advance}$=235 require. DKT ΔFAR is positive on only three of five runs and is not treated as a five-run finding. On XES3G5M, T-KT ΔFAR is negative on all five runs (mean −0.017), and ΔMiss is also negative (mean −0.183): after padding tokens are excluded, sparse KCs do not show a higher miss rate than dense KCs.

Coincidence of digits: T-KT dense E[FAR] at τ=0.7 on seed 42 is 0.113, close to the four-partition dense ECE 0.114. They are different quantities.

Table 6. Gate robustness at τ=0.7 on ASSISTments 2012. Five training runs (seeds 42, 2024, 2025, 2026, 2027) across four unique learner partitions (2025 and 2026 share a split). Partition-level ΔFAR averages seeds 2025 and 2026 first; T-KT mean 0.056, range 0.015–0.087. Mean N, $N_{advance}$, and $N_{incorrect}$ are sparse-stratum denominators. GKT/CL4KT remain seed 42 only.

Model	Mean ΔFAR	SD	Runs ΔFAR>0	Mean N	Mean $N_{advance}$	Mean $N_{incorrect}$
T-KT	0.047	0.033	5/5 runs; 4/4 unique partitions	413	227	155
DKT	0.033	0.048	3/5 runs	413	226	155

D. Dataset-dependent explanatory analysis

If a learning platform uses KT probabilities only to sort students, Table 3 may suffice. If it uses p to skip practice or declare mastery, Table 3 is an insufficient evaluation summary. The ASSISTments T-KT cell is an illustrative case: competitive AUC, Limited-but-estimable sparse occupancy (about 19% of KCs fall below the sparse threshold on fold 0), a monotonic ECE rise, and a gate that yields more higher-error sparse advance decisions. The XES3G5M cell is the contrasting case: plenty of sparse mass and Reliable occupancy, yet T-KT ECE does not degrade. Junyi shows that the protocol can refuse a sparse claim when the bucket is empty.

Table 7 separates two questions. C1–C2 determine whether a sparse contrast can be estimated under the protocol thresholds; C3 and other structural descriptors help characterize the direction of the observed pattern. C1 is a non-empty low-frequency tail under the split actually used; C2 is enough sparse test events for at least a Limited-flag ECE. C3 (frequency–difficulty coupling) is not treated as necessary or sufficient for an adverse gradient. Junyi fails C1–C2. ASSISTments and XES3G5M both meet C1–C2, yet only ASSISTments shows a dense-to-sparse T-KT ECE rise.

Table 7. Empirical conditions associated with the availability and direction of sparse-calibration contrasts on the three evaluated datasets. Sparse mass: share of KCs with train-only frequency f_{train}<100. C3 is not necessary or sufficient for an adverse gradient.

Condition	ASSISTments	Junyi	XES3G5M
A. Estimability			
Sparse mass (C1)	18.9% of KCs	0%	22.5% of KCs
Sparse test support (C2)	415 (L)	empty	1,969 (R)
B. Observed pattern/context			
Difficulty coupling (C3)	ρ=−0.227 (sparse harder)	ρ=−0.416	ρ=+0.110 (weak, inverted)
Item support	median 44.5 (dense 205 vs sparse 1)	median 18 (IQR 5)	median 3 (dense 9 vs sparse 1)
Curriculum-position coupling	ρ=−0.308	ρ=−0.324	ρ=−0.126
Observed T-KT ECE	0.114→0.228	dense→medium only	flat 0.118→0.125

Learner exposure (distinct training learners per KC) enters the regressions below; it is not a C1–C2 estimability criterion. This subsection reports two distinct estimands. Neither is a causal claim.

A test-event-weighted KC-level regression of T-KT expected calibration error (ECE) on those five training-only covariates is a between-KC association: it compares different concepts. ASSISTments (478 KC-fold rows; 261 unique KCs) and Junyi (2,645 KC-fold rows; 1,326 unique KCs) use cluster-robust standard errors at kc_id; XES3G5M is 829 unique KCs in the masked weighted fit. Standardized log(1+ftrain) coefficients are −0.068 [−0.086, −0.050] on ASSISTments, −0.014 [−0.018,

−0.010] on Junyi, and −0.028 [−0.042, −0.014] on XES3G5M in the weighted fit. Difficulty is independently associated on all three datasets. Learner exposure is independently associated on ASSISTments (+0.016 [0.004, 0.027]) and on XES3G5M (+0.013 [+0.005, +0.021]). These signs do not test a frequency dose on a fixed KC.

Within-KC controlled sparsification holds KC identity, the test set, labels, and all other KCs' training rows fixed, and reduces training rows for 30 originally dense KCs (f_{train}≥500; seed 42, fold 0) to 500 or 50 rows. The selection rule was fixed and recorded before reduced-evidence ECE was inspected. Table 8 reports protocol endpoints (500 and 50 rows) for DKT and T-KT on all three datasets. Complete results for all models, datasets, and reduction levels are reported in Supplementary Table S2. Reducing training evidence for the same KC does not universally worsen calibration: several CIs lie below 0 or include 0. XES3G5M T-KT shows a positive ΔECE at 50 rows (+0.110 [+0.071, +0.156]); DKT at 500 rows is also positive (Table 8; Supplementary Table S2). The observational ASSISTments T-KT dense-to-sparse ECE gradient (0.114→0.228) is not reproduced by sparsifying originally dense ASSISTments KCs (T-KT, 50 rows: +0.002 [−0.021, +0.025]). Frequency alone is therefore not a universal causal explanation. Junyi T-KT at 50 rows does show a large positive ΔECE; that cell shows a within-KC increase is possible, not that it is a law.

Table 8. Within-KC controlled sparsification at protocol endpoints (500 and 50 training rows; seed 42; 30 originally dense KCs per dataset). The 100-row level is omitted here by that pre-declared endpoint rule, not by outcome. ΔECE = reduced−full; positive means worse calibration. 95% CIs bootstrap over KCs. Not a causal law for real-world sparsity. Complete results for all models, datasets, and reduction levels are reported in Supplementary Table S2.

Dataset	Model	Reduction	ΔECE	95% CI	Interpretation
ASSISTments 2012	DKT	500 rows	−0.047	[−0.060, −0.033]	ECE lower
ASSISTments 2012	DKT	50 rows	−0.017	[−0.034, +0.002]	CI includes 0
ASSISTments 2012	T-KT	500 rows	−0.026	[−0.047, −0.002]	ECE lower
ASSISTments 2012	T-KT	50 rows	+0.002	[−0.021, +0.025]	CI includes 0
Junyi Academy	DKT	500 rows	−0.021	[−0.041, −0.001]	ECE lower
Junyi Academy	DKT	50 rows	−0.004	[−0.022, +0.013]	CI includes 0
Junyi Academy	T-KT	500 rows	+0.101	[+0.081, +0.120]	ECE higher
Junyi Academy	T-KT	50 rows	+0.135	[+0.110, +0.161]	ECE higher
XES3G5M	DKT	500 rows	+0.142	[+0.104, +0.189]	ECE higher
XES3G5M	DKT	50 rows	+0.161	[+0.127, +0.196]	ECE higher
XES3G5M	T-KT	500 rows	+0.041	[+0.020, +0.063]	ECE higher
XES3G5M	T-KT	50 rows	+0.110	[+0.071, +0.156]	ECE higher

E. Exploratory GKT/CL4KT result

GKT (a train-only graph) and a CL4KT protocol adapter are scored only on ASSISTments 2012 fold 0 (seed 42). They are exploratory, single-fold, ASSISTments-only instantiations: not a state-of-the-art comparison, not a proposed method, and not an official CL4KT checkpoint. Table 5: GKT FAR is 0.205 [0.194, 0.217] dense versus 0.220 [0.149, 0.295] sparse (ΔFAR=+0.015; 95% CI [−0.054, 0.092] includes 0). The CL4KT adapter FAR is 0.185 [0.176, 0.194] versus 0.240 [0.159, 0.330] (ΔFAR 95% CI [−0.018, 0.142] includes 0). A CI that includes 0 is not evidence that either architecture is safer in production.

V. DISCUSSION

A. Main empirical findings

Three empirical regularities, not laws, structure the results. First, training frequency is not a universal predictor of AUC failure: on XES3G5M, sparse-stratum AUC is higher than dense-stratum AUC for DKT and T-KT (Reliable occupancy), and Junyi's learner-based sparse bucket is empty rather than a ranking collapse. Second, calibration vulnerability is dataset-dependent. ASSISTments 2012 T-KT ECE is associated with a dense-to-sparse rise under Limited sparse support (Table 4, N=415); XES3G5M T-KT ECE is a counter-pattern that stays essentially flat despite Reliable sparse occupancy. Calibration is not guaranteed to worsen as frequency falls. Third, threshold behavior can differ by frequency stratum. The simulated ASSISTments gate (Tables 5–6) can raise T-KT FAR on sparse KCs relative to dense KCs, with ΔFAR positive in 5/5 training runs and positive in 4/4 unique partition-level estimates; on XES3G5M, T-KT ΔFAR and ΔMiss are both negative after padding is excluded, so a flat ECE is not by itself a decision-error result.

B. Practical implications for educational technology

Before using one global KT probability threshold for remediation or advancement, four checks are warranted under the evaluated conditions: (i) inspect KC-frequency occupancy on the training fold actually used, including empty buckets; (ii) evaluate per-stratum calibration separately from AUC; (iii) evaluate threshold-error metrics (FAR, expected FAR, Excess FAR, and Miss) with their denominators; (iv) ensure sufficient sample support before treating a slice as actionable. R/L/I flags are descriptive occupancy labels, not inferential guarantees. A population AUC win, or an exploratory GKT/CL4KT run, is not by itself evidence that a global gate is safer on the tail.

C. Why datasets differ

The three logs differ on C1–C2 estimability and on structural context (Table 7). Junyi's learner-based sparse bucket is empty. ASSISTments and XES3G5M both have a non-empty sparse tail and at least Limited test support, yet T-KT ECE rises on ASSISTments and stays essentially flat on XES3G5M. Frequency–difficulty coupling, item support, and curriculum-position coupling are reported as context, not as necessary or sufficient conditions for that direction. Within-KC sparsification (Table 8) does not reproduce the observational ASSISTments T-KT ECE gradient. These are associations on these datasets. Untested explanations—curriculum hierarchy, tagging granularity, ceiling effects, and item semantics—are hypotheses unless experimentally isolated, which this study does not do.

D. Limitations

Next-response correctness is not latent mastery: y=0 is an incorrect next attempt, not a latent-skill diagnosis. The threshold gate is simulated at a locked τ and is not a classroom policy. There is no classroom RCT. GKT and the CL4KT adapter are exploratory, single-fold, ASSISTments-only instantiations, not a state-of-the-art comparison. Temporal evaluation uses a single corrected cutoff (seed 42), not a multi-cutoff variance estimate. Main multi-run summaries use only four unique learner partitions (seeds 2025 and 2026 share a split). R/L/I are descriptive support flags. ECE depends on binning. Three datasets cannot establish a universal diagnostic law.

VI. CONCLUSION

Under the evaluated conditions, KT probabilities that look adequate on aggregate AUC can still be poorly calibrated on sparse KCs in some dataset-model settings, and a global threshold can then raise the false-advance rate on those skills. That association appears for T-KT on ASSISTments 2012 (Limited sparse support). It is not reported for Junyi's empty learner-based sparse bucket, and XES3G5M T-KT ECE remains essentially flat. Sparse-concept occupancy, calibration, and threshold-error checks are therefore conditionally useful for educational-technology gates. This paper is a simulated decision-error check, not a new KT model and not a classroom intervention.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Author 1: Conceptualization, Methodology, Software, Formal analysis, and Writing (led diagnostic design, experiments, and manuscript). Authors 2 and 3: Software (data processing and baseline runs). Author 4: Methodology (methodological review). Author 5: Supervision and Writing (manuscript revision). All authors approved the final version.

ETHICAL STATEMENT

This study is a secondary analysis of publicly released, de-identified benchmark educational datasets (ASSISTments 2012, Junyi Academy, and XES3G5M). No new human participants were recruited and no classroom intervention was conducted. The work uses only records already released by the original dataset providers.

DATA AND CODE AVAILABILITY

Anonymized code, data-preprocessing scripts, and evaluation pipelines are available for peer review at https://anonymous.4open.science/r/Sparse-Concept-and-Calibration-6E5B/. Source benchmark datasets are publicly released by their original providers. A public repository will be released upon acceptance. Named camera-ready manuscripts are not part of this IJIET review snapshot; ignore them if an older copy still contains them.

GENERATIVE AI STATEMENT

During manuscript preparation, the authors used ChatGPT, Claude, and Google Antigravity for language polishing, formatting, consistency checking, and reproducibility prompt preparation. Exact ChatGPT, Claude, and Google Antigravity model identifiers were not retained; later revision used Cursor Grok 4.6. AI was not used to fabricate or alter experimental results. After using these tools, the authors reviewed and edited the content. The authors remain responsible for all content. Generative AI is not listed as a co-author.

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